Power System & Charging Station — Scooter-Class Motor Configuration

Full power circuit design: motors, battery, distribution, wiring and charging dock

Field	Detail
Project	Autonomous AI-Powered Hospital Delivery & Infection-Control Robot
University / Dept.	Benha National University — Mechatronics and Communications Engineering
Subsystem owner	Ziad — Power system and charging station
Revision	Motors upgraded to scooter-class 24V hub motors (supersedes the earlier 12V gearmotor version)

1. Why scooter-class motors change the design

E-scooter wheel hub motors are brushless (BLDC), run at 24V or 36V instead of 12V, and are rated 250–500W continuous — roughly 5–10x the power of the small 12V gearmotors used in the earlier version. That changes four things at once: the battery voltage and capacity, the motor driver type (a BLDC controller instead of a simple H-bridge), the wire gauges and fuse ratings, and the charger. Everything below reflects that.

2. Motor selection

Recommended motor

Two identical 24V brushless hub motors (direct-drive, wheel-integrated), 250W continuous / up to 350W peak each, the same class used on consumer e-scooters. Each has 3-phase windings plus a built-in 5-wire Hall sensor set used both for commutation and as a coarse speed sensor.

Parameter	Value	Why
Voltage	24V nominal	matches a standard 7S Li-ion / 8S LiFePO4 pack; 36V is an option if more speed is needed, but raises every other rating
Power	250W continuous, 350W peak (per motor)	ample torque for a 15–25kg loaded chassis with ramps and door sills
Drive type	Direct hub drive (no external gearbox)	fewer moving parts than a geared motor, standard on e-scooters
Feedback	3x Hall sensors (built in)	used for BLDC commutation; also gives coarse wheel-speed odometry (lower resolution than a quadrature encoder — fuse with IMU if precise localization is needed)
Wheel size	match existing chassis wheel, typically 8–10 inch	keep ground clearance and speed in the target 0.5–1 m/s indoor range

This is a deliberate trade-off: scooter hub motors are more powerful and more available off-the-shelf, but Hall-only feedback gives lower odometry resolution than the encoder-equipped gearmotors in the original design. If Nav2 localization accuracy suffers in testing, an add-on magnetic encoder disk per wheel is a straightforward retrofit.

Motor controller (one per wheel)

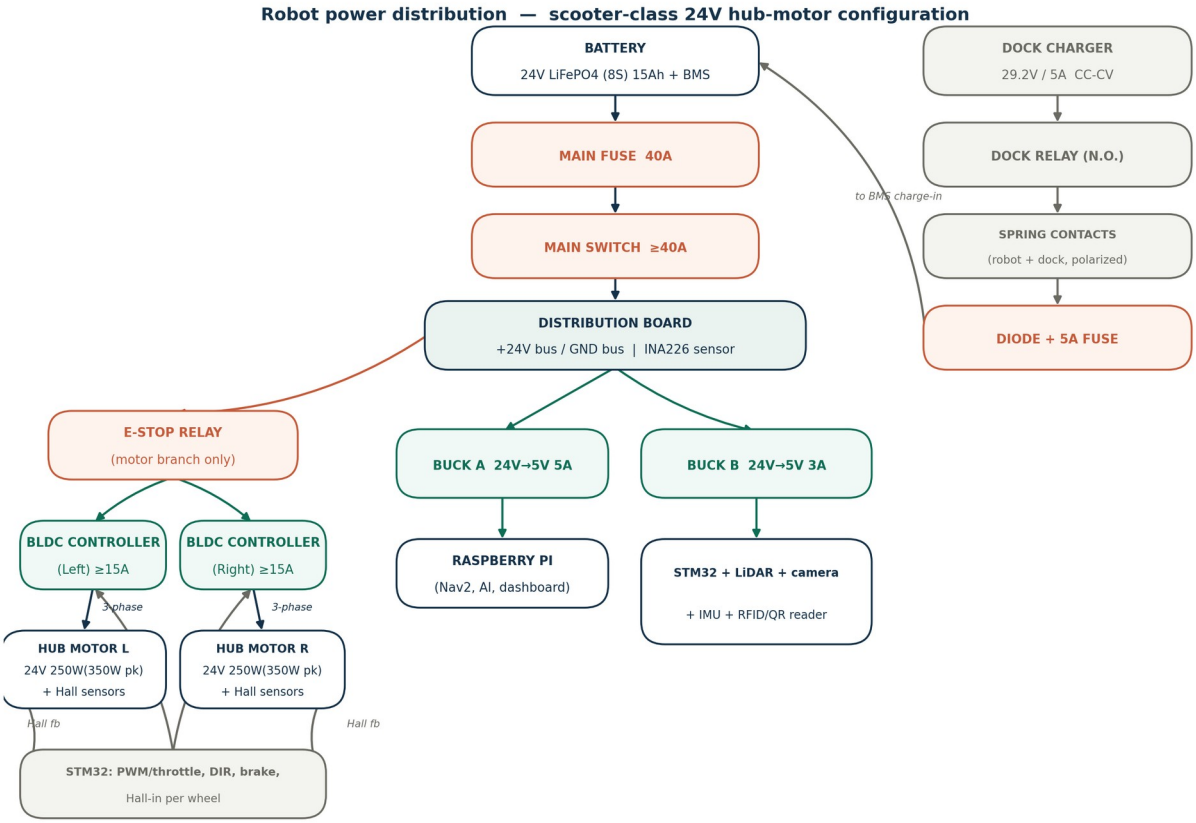
Each hub motor needs a dedicated BLDC controller (the same role a scooter's ESC plays), not a simple H-bridge — it decodes the Hall signals and commutates the 3 phases. Rating: 24V, ≥15A continuous (covers 350W peak / 24V ≈ 14.6A), with Hall sensor input, a PWM or 0–3.3V analog speed-command input, a direction pin, and a brake pin, all driven from the STM32.

3. Battery

Item	Recommended choice
Chemistry	LiFePO4, 8S (24V nominal, 29.2V full-charge) — safer and longer-lived than Li-ion at this power level
Capacity	15 Ah (≈384 Wh) — gives roughly 4–5 hours of mixed driving at an average 700W combined motor + electronics load during active movement, more on standby
BMS	pre-integrated with the pack, continuous rating ≥ 30A, peak ≥ 60A (covers both motors stalling together)
Main fuse	40A, directly at the battery positive terminal
Main switch	automotive disconnect switch, ≥ 40A rated

Note on voltage headroom: an 8S LiFePO4 pack reaches 29.2V at full charge, which exceeds the 26V maximum of the INA219 sensor used in the earlier 12V design. Use an INA226 instead (36V rated) for the main bus voltage/current sensor.

4. Full power circuit diagram



5. Wiring — exact connections

Star-ground topology, as before: every return path goes to one ground bus on the distribution board. Gauges below are sized for the higher current of the scooter motors.

From	To	Wire	Connector	Notes
Battery +	Main fuse (40A)	10 AWG	Ring terminal / XT90	shortest practical run
Main fuse	Main switch	10 AWG	Ring terminal	
Main switch	Distribution board +bus	10 AWG	Screw terminal block	
Battery − (BMS out)	Distribution board GND bus	10 AWG	Ring terminal	
+bus / GND bus	E-stop relay → BLDC controllers	12 AWG	XT60	motor branch only
BLDC controller (L/R)	Hub motor (L/R) phases U/V/W	14 AWG x3	pre-wired motor pigtail connector	do not swap phase order after tuning direction
Hub motor Hall connector	BLDC controller Hall input	22 AWG, 5-wire	JST-XH 5-pin	Vcc/GND/A/B/C
STM32	BLDC controller command input	22 AWG	JST-PH 3-pin (PWM, DIR, EN)	per wheel
+bus / GND bus	Buck converter A (24V→5V 5A)	16 AWG	Screw terminal	Pi rail
Buck A OUT	Raspberry Pi 5V/GND	18 AWG, <10 cm	Soldered pigtail or USB-C PD board	
+bus / GND bus	Buck converter B (24V→5V 3A)	18 AWG	Screw terminal	logic rail
Buck B OUT	STM32, LiDAR, camera, IMU, RFID	22 AWG	JST-PH 2-pin per device	
INA226 sensor	distribution board +bus (in-line) + I2C to STM32/Pi	20 AWG + I2C ribbon	JST-PH + Dupont	36V-rated, matches 29.2V pack voltage
Charging contacts	Diode → 5A fuse → BMS charge input	14 AWG	Spring-loaded pogo pins	polarized shroud, de-energized until docked

6. Charging dock

Stage	Component	Function
1	Dock charger, 29.2V / 5A CC-CV, LiFePO4 profile	matches 8S pack exactly — do not reuse a 24V lead-acid charger, its float voltage is wrong for LiFePO4
2	Dock-side relay (normally open)	contacts stay at 0V until a docked robot is detected
3	Spring-loaded contacts, polarized	robot guided in by a V-shaped mechanical funnel + AprilTag vision alignment
4	Series Schottky diode (≥5A)	blocks reverse current if the robot disengages mid-charge
5	In-line 5A fuse	isolates a contact short from the main pack
6	BMS charge input	BMS performs final CC/CV regulation and cell balancing

Same confirm-before-energize logic as the 12V design: STM32 reads a stable INA226 voltage on the charging line for >300 ms before closing the dock relay, and disables the drive motors for the duration of charging.

7. Safety checklist before first power-on

- Verify battery polarity with a multimeter before connecting to the distribution board.
- Confirm the main fuse (40A) and every branch fuse are rated below their wire's ampacity, not just below the load's rated current.
- Bench-test each BLDC controller with its motor disconnected from the chassis first, to confirm correct Hall wiring and rotation direction before driving wheels.
- Measure the 5V rail under full Pi load before trusting it — target ≥ 5.0V at the Pi header; a 24V-input buck under heavy motor current ripple needs a larger input capacitor than the 12V version did.
- Test the E-stop with the chassis on blocks (wheels off the ground) before any floor test.
- Confirm dock contacts read 0V with a multimeter while undocked, before touching them by hand.

8. Bill of materials (power subsystem)

Item	Qty	Est. unit (EGP)	Est. total (EGP)
24V 250W BLDC hub motor (scooter-class)	2	1,200–1,800	2,400–3,600
BLDC motor controller, 24V ≥15A	2	600–900	1,200–1,800
LiFePO4 24V (8S) 15Ah pack + BMS	1	6,500–9,000	6,500–9,000
Dock charger, 29.2V/5A LiFePO4 CC-CV	1	1,200–1,800	1,200–1,800
Buck converter 24V→5V 5A (Pi rail)	1	180–280	180–280
Buck converter 24V→5V 3A (logic rail)	1	150–220	150–220
INA226 voltage/current sensor	1	150–220	150–220
Main fuse holder + fuses (assorted)	1 set	200–300	200–300
Main switch + E-stop button + relay (40A)	1 set	500–800	500–800
Dock contacts (pogo pins) + diode + connectors	1 set	350–550	350–550
Wiring (10–22 AWG), terminals, heat-shrink	1 set	400–650	400–650
Total (power subsystem only)			13,230–19,200

Estimates are rough local-market figures for scoping only — confirm against actual supplier quotations. The jump from the 12V gearmotor version (~7.7k–11.2k EGP) reflects the higher cost of BLDC hub motors, their controllers, and the larger battery — this is the real cost trade-off of moving to scooter-class motors.